

What is UML ?

Unified model language is being used to visualize and address structural and behavioural requirements of the system.

What are the components of UML?

There is majorly 3 components.

1. Things
2. Relations
3. Diagrams.

What are things ?

These are building blocks of UML.

These are majorly 2 types.

1. Structural.
2. Behavioural

Structural Thing : →

These represents static/physical representation of a system.

Some of the structural things are listed as →

1. class
2. Object
3. Interface
4. use-case
5. Actor
6. Component
7. Node.

Use-case : This notation describes user's goal & Possible interaction with the system.

Actor : An entity which interacts with the systems via usecases.

An actor cannot interact with another actor. It can only interacts with the

systems entities.

For example : In chatting apps two user cannot directly talk they communicate via usecase of the system.

Component: Section of a system, is being made of multiple entities.

Node: This physical representation of component .

example: Kafa Node. [Distributed server]

Behavioral thing :

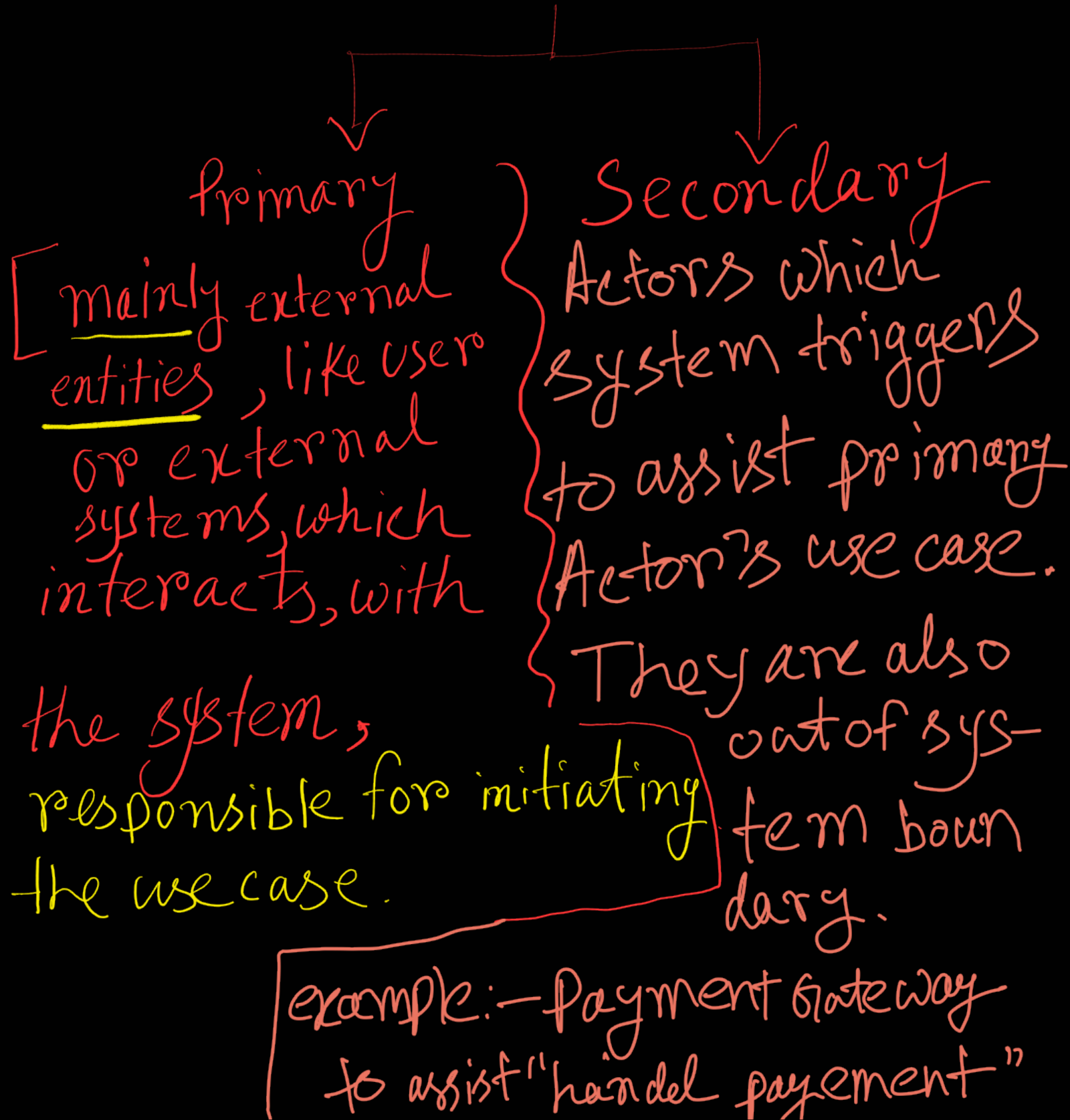
These typically represents dynamic behaviours of the system for different situation .

This can be represent in UML dignam using usecase diagrams , sequence diagrams, Activity diagrams

Use-case diagram:

It mainly consists of Actor's and use case .

Actor



Use case : $\Rightarrow$ It is a single function, performed on a system, by an Actor.

Relationship in a Use-case diagram

1. Association:



2. Generalization-

A parent usecase is generalized by child usecase. It is like inheritance, where Child will inherit all the behaviour of

parent use case. Along with that
.. ..

3. Include : This a relation between two usecases Where base usecase includes another usecase which will assist base usecase to complete the goal of actor.

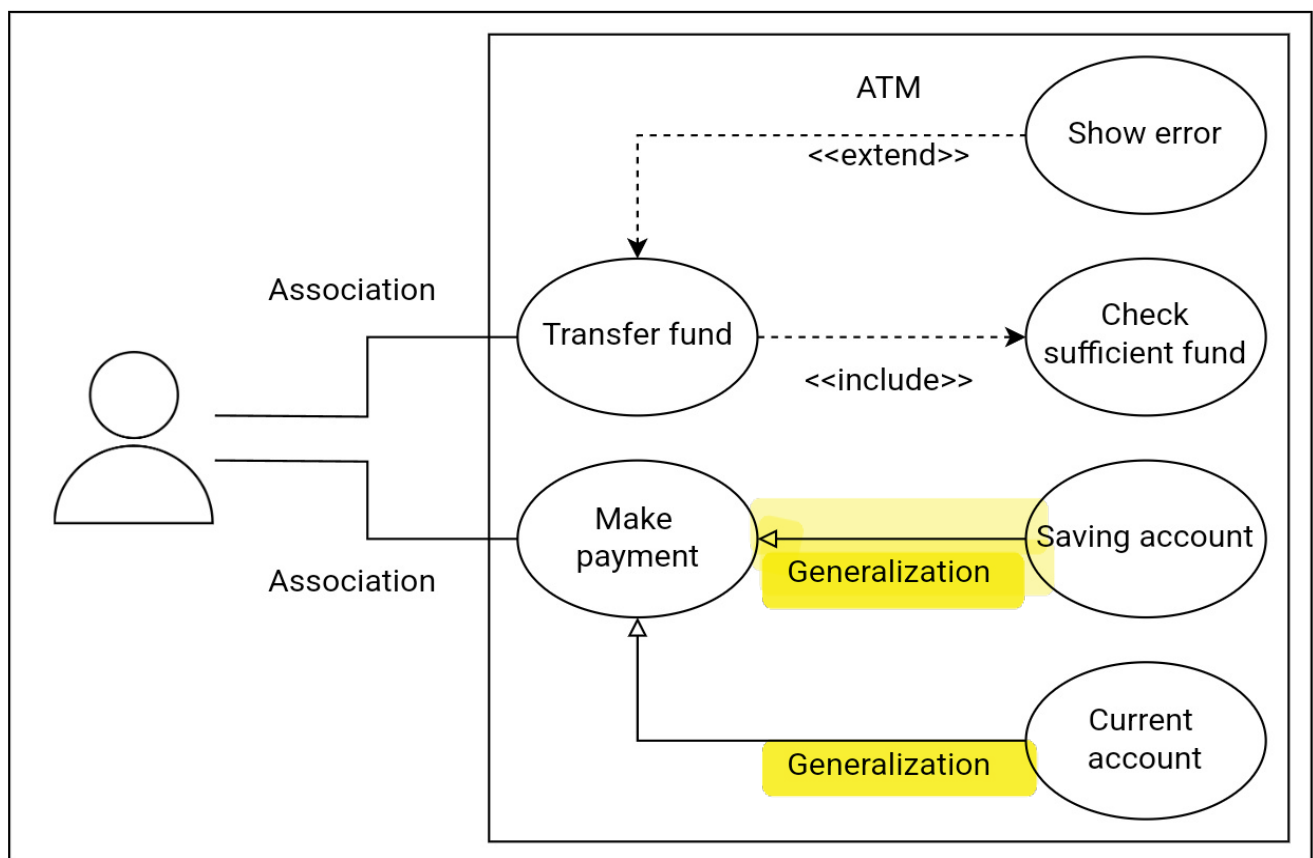
Example: Transferfund use case will trigger another usecase ' isValidAmount(amount).

4. **Extend** : This a relationship between a base Use case & another usecase, where base case will only execute the extende usecase if a particular situation occurs . Otherwise it will not be triggered.

Example: while doing the transferfund if any error occurs then "handelErpor" use case will occur.

base use case

extended.



The use case diagram of an ATM

Use case diagram summary :

1. use case diagram is used to understand Who interacts with the system .
2. what is the system's responsibilities (identifying Use-cases)
3. What are the external dependencies system required.
4. How does actors and use cases are connected.

Pillars of usecase diagrams are

1. use case
 2. Actor
- &The relationship between them .
Association, Generalization , include, extend (refer earlier)

Advantages of usecase diagram:-

1. It helps us to understand, what are

the external entities & systems will interact with the system.

2. It helps to identify dependencies .

3. It helps to understand how actor's will interact with the system, and to complete the use case what dependencies & it's use case system will have to include.

4. It helps technical & non technical personnel to understand the system's architecture & dependencies.

which helps them to understand & meet bussiness requirements.